valence electrons between these ions. Thus, a partial covalent character is developed between them. This phenomenon is called polarisation.

The ability of a cation to polarise an anion is called its polarising ability and the tendency of an anion to get polarised is called its polarisability. The extent of polarisation in an ionic compound is given by the Fajans rules

### Fajans Rules

(i) To show greater covalent character, both the cation and anion should have high charge on them. Higher the positive charge on the cation, greater will be the attraction on the electron cloud of the anion. Similarly higher the magnitude of negative charge on the anion, greater is its polarisability. Hence, the increase in charge on cation or in anion increases the covalent character

Let us consider three ionic compounds aluminum chloride, magnesium chloride and sodium chloride. Since the charge of the cation increase in the order $Na^+ < Mg^{2+} < Al^{3+}$, the covalent character also follows the same order $NaCl < MgCl_2 < AlCl_3$.

(ii) The smaller cation and larger anion show greater covalent character due to the greater extent of polarisation.

Lithium chloride is more covalent than sodium chloride. The size of $Li^+$ is smaller than $Na^+$ and hence the polarising power of $Li^+$ is more. Lithium iodide is more covalent than lithium chloride as the size of $I^-$ is larger than the $Cl^-$. Hence $I^-$ will be more polarised than $Cl^-$ by the cation, $Li^+$ .

(iii) Cations having $ns^2$ $np^6$ $nd^{10}$ configuration show greater polarising power than the cations with $ns^2$ $np^6$ configuration. Hence, they show greater covalent character.

CuCl is more covalent than NaCl. Compared to $Na^+$ (1.13 Å) . $Cu^+$ (0.6 Å) is small and have $3s^2$ $3p^6$ $3d^{10}$ configuration.

Electronic configuration of $Cu^+$

$[Ar]\ 3d^{10}$

Electronic Configuration of $Na^+$

$[He]\ 2s^2, 2p^6$

## 10.6 Valence Shell Electron Pair Repulsion (VSEPR) theory

Lewis concept of structure of molecules deals with the relative position of atoms in the molecules and sharing of electron pairs between them. However, we cannot predict the shape of the molecule using Lewis concept. Lewis theory in combination with VSEPR theory will be useful in predicting the shape of molecules.

**Important principles of VSEPR Theory are as follows:**

1. The shape of the molecules depends on the number of valence shell electron pair around the central atom.
2. There are two types of electron pairs namely bond pairs and lone pairs. The bond pair of electrons are those shared between two atoms, while the lone pairs are the valence electron pairs that are not involved in bonding.
3. Each pair of valence electrons around the central atom repels each other and hence, they are located as far away as possible in three dimensional space to minimize the repulsion between them.



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